Interface Board

User Guide

SUMMARY: The Atmosic Interface Board provides a host computer access to some commonly needed development and debug interfaces on an Atmosic wireless SoC based Evaluation board or other target hardware.

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1. Overview

The Atmosic Interface Board provides a host computer access to some commonly needed interfaces on an Atmosic chip based DUT, such as

- Serial Wire Debug (SWD)
- Debug UART
- 4-wire HCI UART or 2-wire for Direct Test Mode (DTM)
- Reset/Powerdown (PWD) and Boot Mode signals

2. Interface Board Description

Figures 1 and Figure 2 show the main components on version 011-01 (V2) and 011-02 (V3), respectively, of the Interface Board.

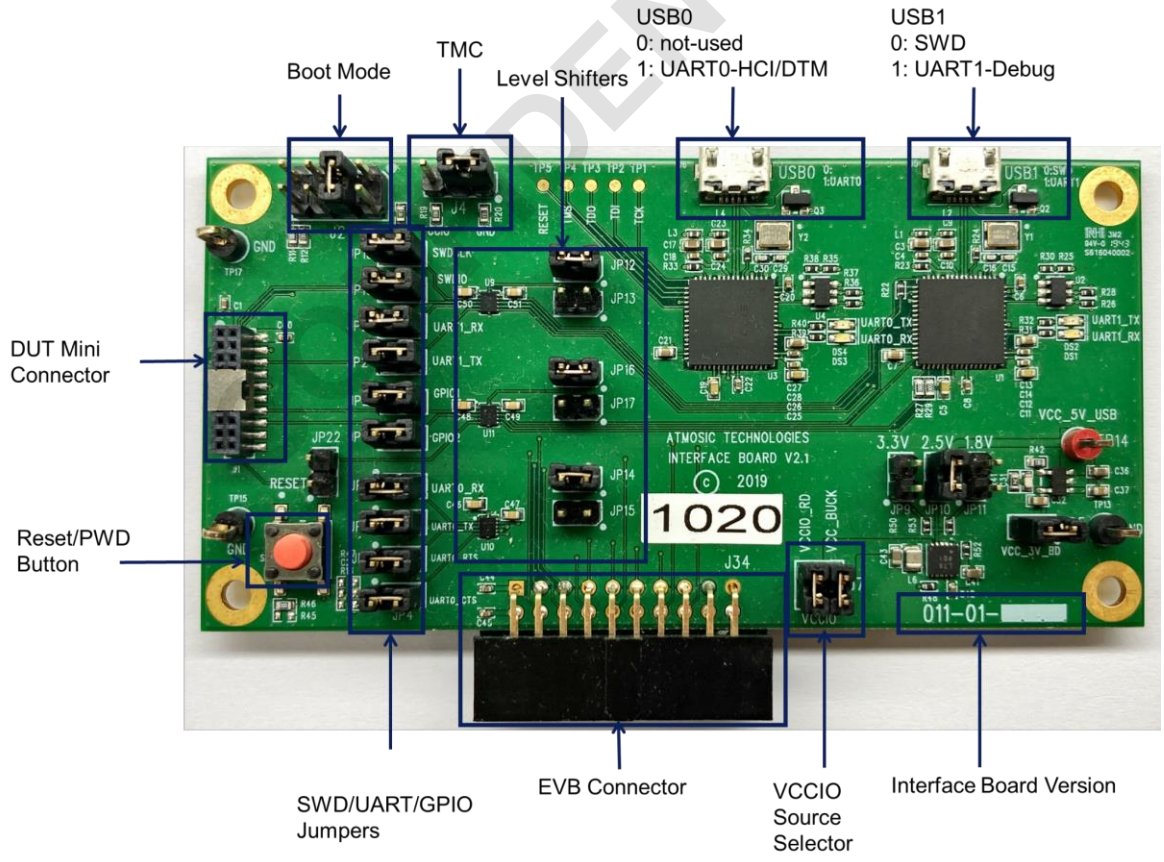


*Figure 1 - Interface Board Version 011-01 (**V2**)*

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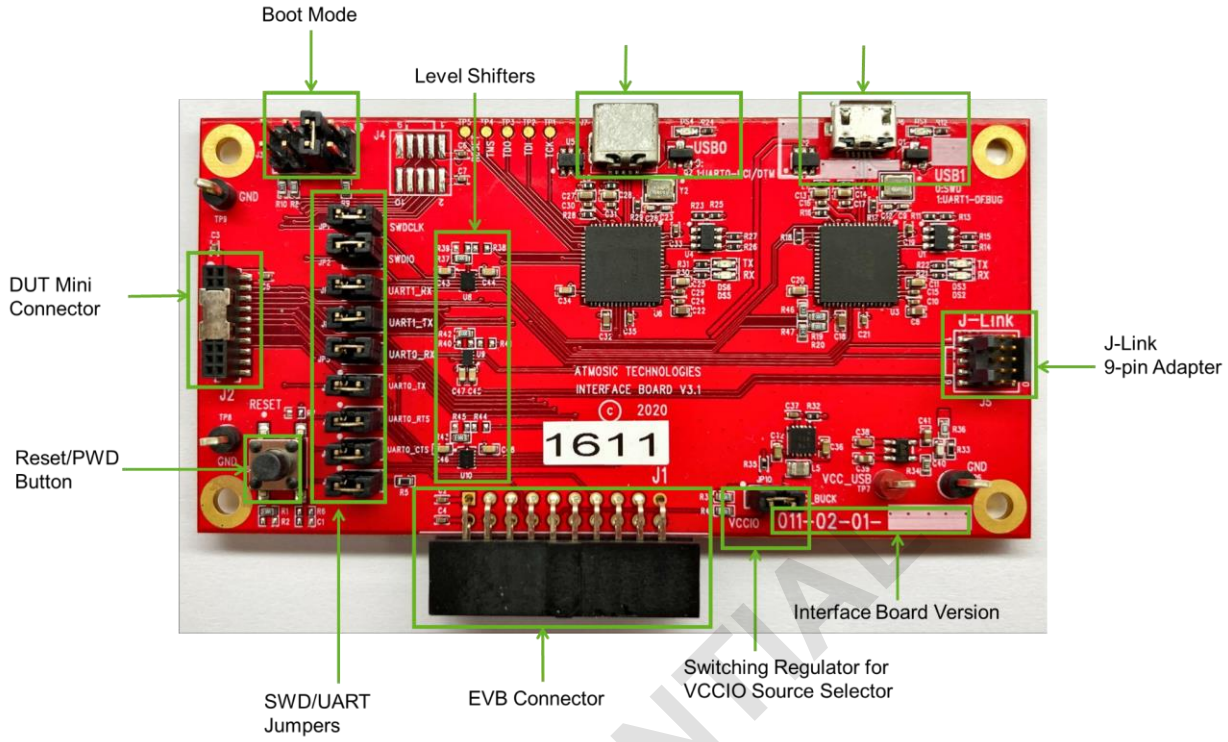


*Figure 2 - Interface Board Version 011-02 (**V3**)*

The latest version V4 is shown in Figure 3, it is modified from V3, and the main changes are described in the following.

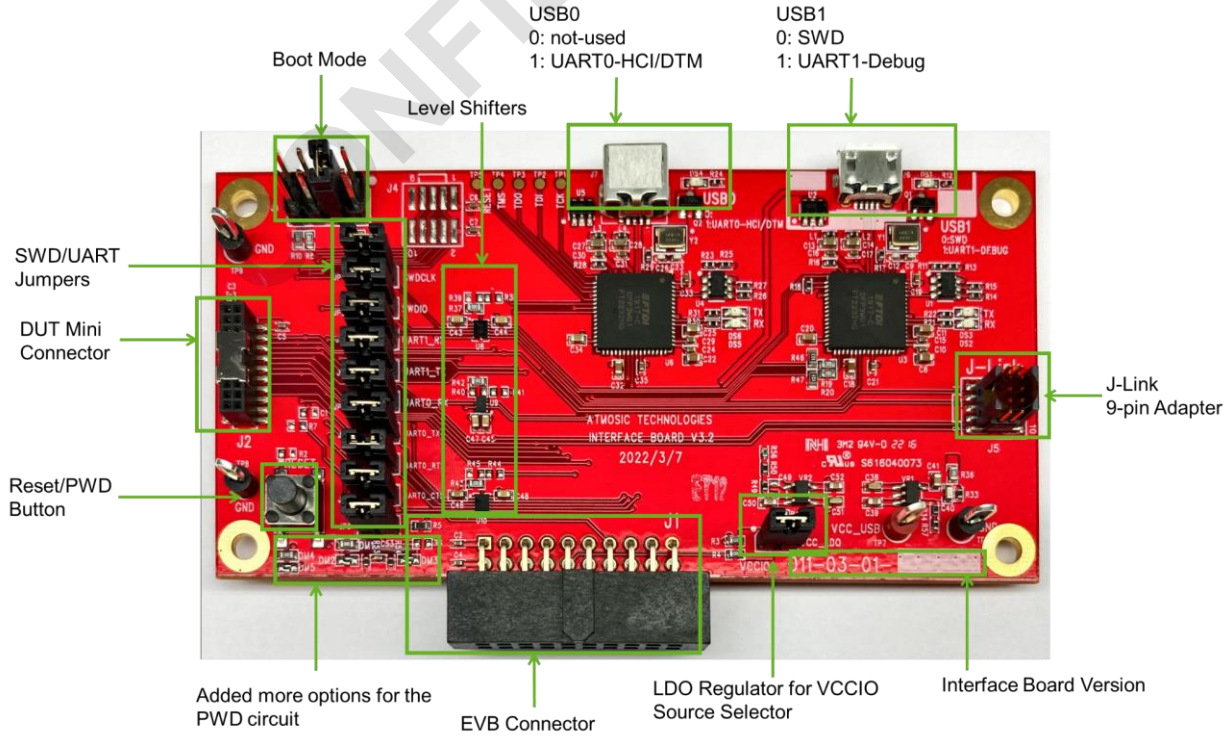


*Figure 3 - Interface Board Version 011-03 (**V4**)*

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- Interface Board Version: The bottom right corner shows the version of the Interface Board. Figure 1 is an example of 011-01 (V2), Figure 2 is an example of 011-02 (V3.1), and Figure 3 is an example of 011-03 (V4). The differences between V3 and V4 are
    - replaced the switching regulator in V3.1 by LDO regulator in V4, for the VCCIO Source Selector.
    - added more options for the PWD circuit for V4.
- USB0: This is a micro-USB port on V2 (mini-USB port on V3) that is connected to an FTDI FT2232H, which provides two serial ports. This should show up as "Atmosic RDI USB0" in Windows Device Manager, and /dev/ttyUSBxxx in Linux.
    - Port 0: Reserved for internal use
    - Port 1: 4-wire HCI UART or 2-wire DTM to the Atmosic chip
    - On V3, an LED (DS4) will be solid blue when USB0 is connected to power
- USB1: This is a micro-USB port that is connected to an FTDI FT2232H, which provides two serial ports. This should show up as "Atmosic RD[1]I USB1" port in Windows Device Manager, and /dev/ttyUSBxxx in Linux.
    - Port 0: SWD
    - Port 1: Debug UART
    - On V3, an LED (DS1) will be solid blue when USB1 is connected to power
- Boot Mode - Benign Boot, pull high to keep CPU in an idle state during boot.
    - Pins 1-2: Benign Boot
    - Pins 3-4: USB1 FTDI control for use with SDK (Default)
    - Pins 5-6: Reserved for internal use
    - Pins 7-8: Normal Boot
- Board Connectors: There are two options to connect the Interface Board to an Atmosic board:
    - Mini Connector: This is a 1.27 mm pitch 2x10 20-pin Female interface.

[1] Reference Design Interface

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- EVB Connector: This is a 2.54 mm pitch 2x20 20-pin Female interface, side launch designed to connect directly to an Evaluation Board (EVB). See Figure 4 and Figure 5 for pin definition.
  - V2 Pin Definition:

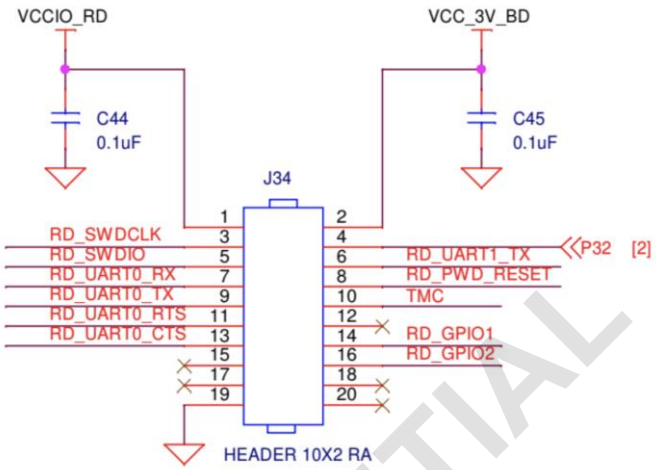


*Figure 4 - **V2** 2x10 20-pin Connector Definition*

  - **V3** Pin Definition:

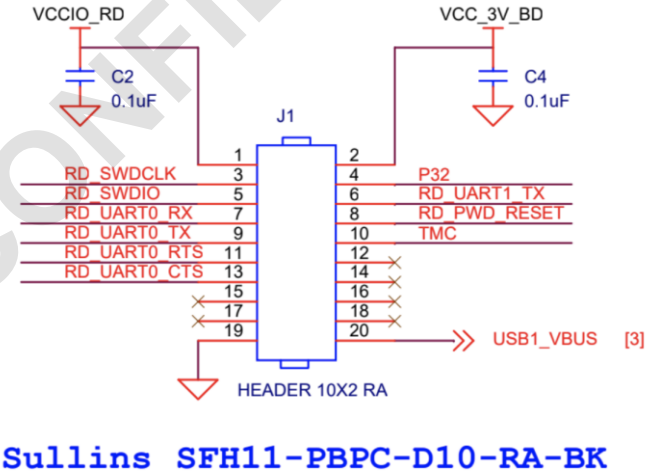


*Figure 5 - **V3** 2x10 20-pin Connector Definition*

- VCCIO Source Selector: These jumpers allow the user to either drive or use the IO supply on the Atmosic board. There are 2 valid modes determined by J7 pins 1-3 on V2 and J10 on V3:
  - Shorted: VCCIO is supplied by the Interface board to the Atmosic board (Default)

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- Open: VCCIO is supplied by the Atmosic board

J7 pins 2-4 should always be shorted on V2 and have been replaced with a 0 Ω resistor on V3.

- Level Shifters: These are level shifters to provide voltage isolation between the FTDI FT2232H and the Atmosic chip. There are a total of 3 level shifters, each with its own output enable and disable jumpers.
  - Level Shifter 1 is for SWD and Debug UART
  - Level Shifter 2 is for other signals from the USB1 FTDI chip
  - Level Shifter 3 is for HCI UART/DTM
- SWD/UART/GPIO Jumpers: These jumpers allow the user to disconnect individual signals between the level shifters and the board connectors.
- Reset/PWD Button & Jumper: Used to Reset/PWD the Atmosic chip. The jumper is loaded by default for SDK use as it allows control from the USB1 FTDI chip. On V2, JP22 must be open to use the button.
- J-Link 9-pin Adapter (only on V3): This is a 1.27 mm 9-pin male Samtec FTSH connector that works with the J-Link 9-pin Cortex-M Adapter.

Figure 6 shows the pin definition:

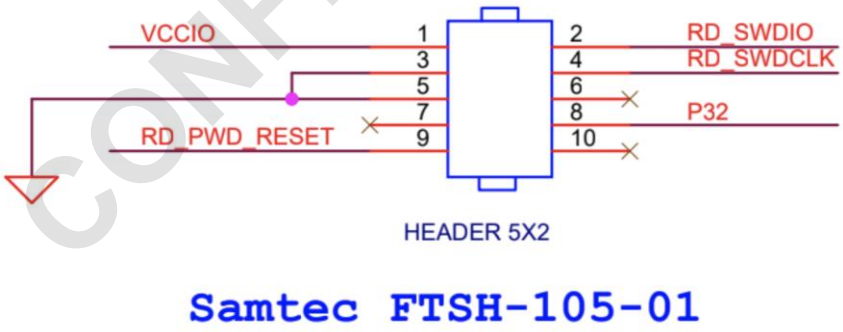


Figure 6 - J-Link Connector Definition

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Appendix - Frequently Asked Questions

1) How to supply power to the Interface Board?
   There are a few options to power the Interface Board. These options will provide power to both Bucks and the two FTDI chips.

   a) Connect a USB cable to a host computer (default and recommended method). Connecting either USB0 or USB1 will supply the necessary power.
   b) Connect a 5 V supply at the red loop and ground to any black loop
   c) Connect a USB cable to a USB power adapter.

2) Extra currents are seen during power measurements when the Interface Board is connected:
   When performing power consumption measurements, sometimes unexpected VCCIO current may be seen. This is due to the weak drive of the bi-directional level shifters used on the board, which have an effective 4 KΩ output resistance.

   Disconnecting the various jumpers (SWD/UART) can help isolate which specific line is causing the issue. Leave the jumper disconnected when it is identified.

3) Debug UART lines show garbage characters:
   Sometimes the UART lines may not be pulled correctly due to the level shifter's bi-directional nature, which has a 4 KΩ output resistance. This may sometimes cause the initial set of characters to be shown as corrupted.

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Revision History

Date	Version	Description
June 13, 2022	0.52	Updated Interface Board Description section.
April 14, 2021	0.51	Updated format, no content change.
July 24, 2020	0.50	Initial version created.

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